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SKILL WALLET
A SmartBridge Product

Trailblazer WORKFORCE PARTNER

SALESFORCE

SOQL and SOSL

Querying Salesforce data: structure, relationships, text search and bulk-safe Apex

Welcome. Today you will learn to choose the right query tool and write it safely in Apex.

```
SELECT Id, Name, Industry
FROM Account
WHERE Industry = 'Technology'
ORDER BY Name ASC
LIMIT 10
```

Start with the question, not the query

Which accounts have open opportunities?	**SOQL**	Known object, exact filter
Which records mention Acme, anywhere?	**SOSL**	Text search across objects
How many contacts does each account have?	**SOQL**	Aggregates need GROUP BY

The data model



Sketch the model first. A dot goes up a lookup; a subquery goes down a list.

Write SOQL: clauses first, then relationships

```
SELECT Id, Name, Industry
FROM Account
WHERE Industry = 'Technology' AND AnnualRevenue > 1000000
ORDER BY Name ASC
LIMIT 10
```

Child to parent: dot notation

```
SELECT Name, Account.Name
FROM Contact
```

Each Contact returns its parent's name in one query.

Parent to child: subquery

```
SELECT Name, (SELECT LastName FROM Contacts)
FROM Account
```

Each Account returns a list of its Contacts.

Filter, group and join across records

Filters

= != < > <= >= compare values. LIKE 'Acme%' matches a pattern. Date literals like LAST_N_DAYS:30. Write Field = null.

```
List<Account> a = [SELECT Id
  FROM Account WHERE Name =
:n];
```

Aggregates

COUNT per group with GROUP BY. HAVING keeps only groups that pass. Read rows with get('total').

```
SELECT AccountId, COUNT(Id)
total
FROM Contact
GROUP BY AccountId
HAVING COUNT(Id) > 5
```

Semi- and anti-joins

IN keeps accounts that have a match. NOT IN keeps accounts with none. One query, no loop.

```
SELECT Id, Name FROM Account
WHERE Id NOT IN
  (SELECT AccountId FROM
Opportunity)
```

SOQL for exact data, SOSL for text

Aspect	SOQL	SOSL
Purpose	Exact, structured retrieval	Text search across many objects
Basic form	SELECT … FROM … WHERE	FIND {term} IN ALL FIELDS RETURNING …
Filtering	=, !=, LIKE, IN, date literals	Whole words, with * and ? wildcards
Relationships	Dot notation and subqueries	Not used to join records
Aggregates	COUNT, SUM, GROUP BY, HAVING	Not supported
Sync limit	100 queries per transaction	20 queries per transaction

SOSL syntax

```
FIND {Acme} IN NAME FIELDS
RETURNING Account(Id, Name),
  Contact(FirstName, LastName, Email)
```

Default to SOQL

Reach for SOSL only when the user is searching by text, such as a name typed into a search box.

Bulk-safe Apex, and SOSL in code

Avoid: a query inside a loop

```
for (Contact c : contacts) {
  Account a = [SELECT Name FROM Account
    WHERE Id = :c.AccountId];
}
```

Prefer: collect the Ids, query once

```
Set<Id> ids = new Set<Id>();
for (Contact c : contacts) ids.add(c.AccountId);
Map<Id, Account> accs = new Map<Id, Account>(
  [SELECT Id, Name FROM Account WHERE Id IN :ids]);
```

SOSL in Apex: one list per object, in RETURNING order

```
List<List<SObject>> results = [FIND {Acme*} IN ALL FIELDS
  RETURNING Account(Id, Name ORDER BY Name LIMIT 5),
  Contact(Id, Name, AccountId) LIMIT 5];
List<Account> accs = (List<Account>) results[0];
```

Know the limits, and avoid four mistakes

100 / 200	50,000	20	2,000
SOQL queries per transaction, sync / async	Rows retrieved by SOQL	SOSL queries per transaction	Records returned by SOSL

1

Query in a loop

Collect Ids, query once with IN, read from a Map.

2

No empty-set check

Skip the query when there are no Ids.

3

SOSL for exact filters

Use SOQL WHERE for fixed values such as Status.

4

Unbounded queries

Select needed fields, filter tightly, add LIMIT.

Knowledge check and hands-on lab

1. Which query returns each Contact's parent Account name?
A) SELECT Name, Account.Name FROM Contact
B) SELECT Name, (SELECT Name FROM Account) FROM Contact
C) SELECT Name FROM Account WHERE Contact.Name != null

2. Why does HAVING COUNT(Id) > 5 come after GROUP BY?
A) WHERE cannot filter on an aggregate such as COUNT
B) HAVING runs before rows are grouped
C) HAVING is faster on large data sets

3. Which Apex filter is safe from SOQL injection?
A) WHERE Name = '" + name + "'
B) WHERE Name = :name
C) WHERE Name LIKE name

Hands-on lab (25 min, in pairs)

- Query: Technology accounts with revenue above 1,000,000, sorted by name, top 10.
- Anti-join: contacts whose account has no opportunities.

Show your query, the row count and one thing you would change.

Key takeaways and next steps

1. **SOQL** returns exact, structured data: fields, filters, relationships and aggregates.
2. **Dot notation** goes up a lookup; a subquery goes down to a list.
3. **Bind variables and bulk queries** keep Apex safe and within limits.
4. **SOSL** searches text across many objects and returns one list per object.
5. **Default to SOQL.** Choose SOSL only when the user is searching by text.

Resources

- Salesforce Developers: SOQL and SOSL reference
- Apex Developer Guide: Execution governors and limits
- Trailhead: Apex and SOQL modules

Feedback

What was clearest? Which example would you reuse? What should we change next session? Submit via the SmartBridge feedback form.